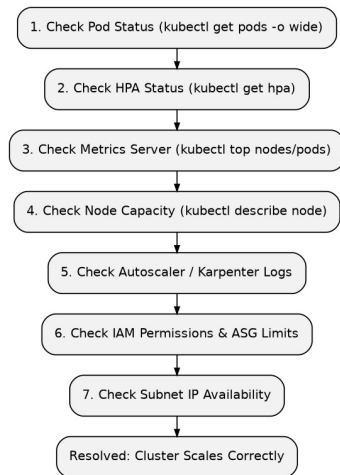


EKS Scaling Troubleshooting Assignment

Assignment: Troubleshooting EKS Cluster Scaling Issues

Question

An EKS cluster is not scaling up properly. What troubleshooting steps would you take to identify and resolve the issue?



EKS Scaling Troubleshooting Steps

Steps

1. **Pod status** — `kubectl get pods -o wide`. Look for pods stuck Pending.
2. **HPA status** — `kubectl get hpa`. Confirm it's requesting more replicas.
3. **Metrics Server** — `kubectl top nodes/pods`. No data = HPA has nothing to scale on.
4. **Node capacity** — `kubectl describe node`. Look for Insufficient cpu/memory events.
5. **Autoscaler/Karpenter logs** — `kubectl logs -n kube-system deployment/cluster-autoscaler`.
6. **IAM & ASG limits** — Confirm autoscaler IAM role permissions and that ASG **max size** isn't already reached.
7. **Subnet IPs** — Check the subnet hasn't run out of free IP addresses for new nodes.

Common Root Causes

- ASG max size reached; missing IAM permissions for the autoscaler.
- Metrics Server down, so HPA has no data to act on.
- Subnet IP exhaustion blocking new node launches.
- Pod resource requests too large for any available instance type.
- Taints/tolerations or node selectors blocking scheduling.

Resolution

Fix the identified constraint (raise ASG limit, correct IAM role, free subnet IPs, or repair HPA metrics), then confirm with `kubectl get pods` and `kubectl get nodes` that new pods schedule and new nodes join successfully.